

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 3

Date: 22/12/2025

Lab Assignment: 03

Aim:

To write a Python program to calculate the distance between two points using their coordinates.

Objective:

- To understand the mathematical concept of calculating the distance between two points in a 2D coordinate system.
- To implement the distance formula using Python programming.
- To use the math module and its sqrt() function for mathematical calculations.
- To practice taking numerical input from the user and displaying the computed result accurately.

Theory:

The distance between two points (x1,y1) and (x2,y2) is calculated using the formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Python provides the math module which contains the sqrt() function to calculate square roots.

The round() function is used to display the result up to a required number of decimal places.

Algorithm:

1. Start the program.
2. Import the math module.
3. Input the coordinates of the first point (x1,y1).
4. Input the coordinates of the second point (x2,y2).
5. Apply the distance formula.
6. Display the distance between the two points.
7. Stop the program.

Pseudo Code:

```
START
IMPORT math
READ x1, y1
READ x2, y2
distance ← sqrt((x2 - x1)2 + (y2 - y1)2)
DISPLAY distance
```

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END

Code:

```
import math;
print("Enter coordinates of 1st point");
x1=float(input("x1: "));
y1=float(input("y1: "));
print("Enter coordinates of 2nd point");
x2=float(input("x2: "));
y2=float(input("y2: "));
d= math.sqrt((x2-x1)**2+(y2-y1)**2);
print("Distance between the two points: ",round(d,3));
```

Output:

```
Enter coordinates of 1st point
x1:  3.5
y1:  9
Enter coordinates of 2nd point
x2:  -11
y2:  -1
Distance between the two points:  17.614
```

Conclusion:

The program successfully calculates the distance between two points using their coordinates by applying the distance formula and Python's math module.